

Android devices have surpassed Symbian, iOS and BlackBerry devices not only in numbers, but also in terms of features and ease of use. Today, Android-based smartphones are being used in almost every organisation and at every level. This fact alone makes Android a likely target for malware aimed at stealing sensitive information. Google and other device manufacturers are putting in their best efforts to make their devices more and more secure, but a curious mind will always find that one loophole, which will compromise the security of the device. Every day, new malware for Android is being unleashed in the wild (interested readers can read 'Android Malware: The Troublesome Ten' published in the February 2013 edition of *Open Source For You*). While only a few of them have the potential to make it to the headlines, even the lesser ones can bite hard at times.

When it comes to mobile devices, malware is not the only thing to worry about. Theft of mobile devices is another concern for users. According to a survey conducted by Symantec, 53 per cent of Indians have been victims of mobile device theft. This is worrying because of the data stored on the mobile device, which may be personal, corporate or both. Once the device is stolen or lost, that data is at risk.

Besides these, there are other risks such as the mobile device being used by other people, misplaced devices, etc, which are common in day-to-day life. A little precaution goes a long way in mitigating these risks.

Checklist for a secure Android device

There are many ways to make sure your Android device is safe and secure. A few of these are listed here.

Implementing basic security (screen security): Every mobile device provides built-in security features. To start with, users must set a screen lock for their device. Android provides interesting ways of doing this. Users can choose from the standard device PIN, password, pattern lock, face unlock and swipe (swipe is the least secure method). Screen lock can be set from *Settings > Screen Security > Screen Lock*. Users should also fill the *Owner Info* form and set it to display on the lock screen. This will help in retrieving the device, in case it gets lost.

SIM PIN: Users should also consider setting up a SIM PIN. With this option enabled, the device will ask for the SIM PIN every time it boots up. This is very handy in preventing misuse of the SIM in case the device is lost or stolen. With this enabled, the SIM will ask for a PIN if it is inserted into any other device.

Passwords: Passwords should not be made visible except if really required. Users should make sure that this feature remains disabled at all times.

Encryption: Android 2.3.4 and later versions support device encryption. Though this is a must-have security feature, most device manufacturers choose to ignore it in their customised Android version. In case this feature is not available, people can use external apps such as Secret Space Encryptor (SSE), Encryption Manager, Droid Crypt, etc.

Multiple user accounts: Tablets running on Android 4.2+ (Jellybean) provide an option to create separate user profiles for different users. This is helpful if the device is being shared by multiple family members. This feature can be found under *Settings > Device > Users*.

Device administration: This section provides two features, 'Device Administrators' and 'Unknown Sources'. 'Device Administrators' are applications that can control certain features of the device. For example, when an anti-virus app is installed, it becomes a device administrator. It can erase all data on the device, set password rules, etc, on the user's behalf. Don't sweat. There is no malicious intent in doing this. Often, these apps provide an option for users to perform these tasks remotely and, hence, they need to act as device administrators.

The 'Unknown Sources' feature allows the user to side load apps or install apps provided by a third party app store. Apps installed via these modes might contain some malicious code, which could pose a risk to the user's data. This option should remain disabled at all times except for development purposes.

Mobile security suites: Installing a mobile security suite is not only a must and a best practice for securing PCs but also for mobile devices. All the leading anti-virus companies provide a mobile version of their product. A mobile security suite provides a wide variety of features along with an anti-virus. For example, users can trace their phones, lock their devices, etc.

Device back-up: Users should regularly back up the data on their device. By default, Android provides an option to back up the configurations and settings. This is available under *Settings > Personal > Backup & Reset*. This is useful when a device is reset to factory settings, and users want to restore the apps and settings without much trouble. However, it does not back up the data stored on the device. Device data can be backed up either by using the desktop software suite for the device or via third party apps installed on the mobile device. Carbon, Titanium Backup, G Cloud Backup, etc, are some apps that can be used for this purpose. Some of these apps provide an option to store the back-up data on the cloud.

Mobile device management: The increasing trend of Bring Your Own Device (BYOD) has led organisations to hunt for Mobile Device Management (MDM) solutions. BYOD comes with risks like corporate data leakage, corporate data theft, etc. To counter such risks, organisations are deploying MDM solutions in their environment. Users should check with the IT department of their company if any such solution is deployed. MDM not only provides features to protect corporate data but also enables employees to locate and wipe their device in case of theft or loss. Users of Samsung's Android-based devices can use Samsung SAFE for this purpose. Besides this, Samsung has recently launched Samsung Knox under the Samsung SAFE umbrella to provide two environments (corporate and personal) on one device.

Application permissions: A good thing about Android's security architecture is that no application is allowed to access device resources without the consent of the user. This is the